

Name: Mewada Maitrri Manishkumar

Topic: Linux Boot Process, GRUB2, System Logs, Boot Troubleshooting, and System Recovery (RHEL 9)

1. Reboot your CentOS 7 or RHEL 8 virtual machine and, during startup, press the appropriate key to access the GRUB2 boot menu; take a screenshot of the GRUB menu screen.

⇒ Step 1: Reboot the VM

`sudo reboot`

Step 2: Open the GRUB menu

While the VM is starting:

BIOS systems: Press Esc repeatedly (sometimes Shift also works).

UEFI systems: Press Esc repeatedly immediately after the firmware

2. Use the 'journalctl -b' command to view the system logs from the current boot and identify the timestamp when the multi-user.target was reached.

Hint: Search for 'Reached target Multi-User System' in the output.

⇒ Step 1: Display logs from the current boot

`journalctl -b`

Step 2: Search for the target

Instead of scrolling through the entire log, use:

`journalctl -b | grep "Reached target Multi-User"`

3. Simulate a boot failure by editing the /etc/fstab file and adding a fake mount entry, then reboot your VM and document the error

message that appears during boot.

Constraint: Restore your original /etc/fstab after testing to avoid permanent boot issues.

⇒ Step 1: Back up the original file

```
sudo cp /etc/fstab /etc/fstab.backup
```

Step 2: Edit /etc/fstab

```
sudo vi /etc/fstab
```

Add this fake entry at the end:

```
/dev/fakepartition /fake ext4 defaults 0 0
```

Save and exit.

Step 3: Reboot

```
sudo reboot
```

Step 4: Expected boot error

You may see messages similar to:

Failed to mount /fake.

Dependency failed for Local File Systems.

Entering emergency mode.

Or

Give root password for maintenance

Step 5: Recover

At the emergency shell:

Remount the root filesystem as read-write:

```
mount -o remount,rw /
```

Restore the backup:

```
cp /etc/fstab.backup /etc/fstab
```

Reboot:

```
reboot
```

4. Use the 'systemctl list-units --failed' command after booting to identify any failed services, and write down the name of one failed service and the command you would use to check its status.

⇒ List failed services:

systemctl list-units --failed

Example:

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
example.service	loaded	failed	failed	Example Service

Check the service status:

systemctl status example.service

If no services have failed, the output may be:

0 loaded units listed.

5. Suppose your system is stuck at the emergency shell prompt during boot. Use ChatGPT to generate a step-by-step recovery plan for this scenario, then try following those steps on your VM and note any differences or extra steps you had to take.

⇒ Boot the system until the emergency shell appears.

Remount the root filesystem with write access:

mount -o remount,rw /

Check /etc/fstab for incorrect entries:

vi /etc/fstab

Remove or correct the invalid mount entry

Save the file.

Check the filesystem (replace the device with the correct one if needed):

fsck -y /dev/sda2

Reload systemd configuration:

systemctl daemon-reload

Reboot the system:

Reboot